



Midterm Exam-b

Course Title	General Physics I	Course Code	30201101
Semester	1 st	Academic Year	2023-2024
Exam Date	29/11/2023	Day	Wednesday
		Time	14:30-15:20

Student Name		Student ID No.		Section	
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GENERAL INSTRUCTIONS

- Students are not allowed to leave the exam room during the first 30 minutes of the exam.
- Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
- Students must use a blue or black pen. Pencils are not allowed.
- Consumption of any food or beverage is prohibited except in medical cases.
- All electronic devices (e.g., cell phones) are prohibited.
- All cell phones must be turned off and placed on the invigilator's desk in the exam room.
- All notes and reading materials must be placed on the invigilator's desk in the exam room.
- Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
- Write the full information required on this page before commencing your exam.
- Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
- Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** _____

Number of Questions	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
10	Multiple Choices Questions	CLIO,1→10	20(100%)			
5	Solving Questions	CLIO,1→10	10(100%)			
Total			30(100%)			----- /30

Moderation Review			
Comments			
Marks Reviewed by			Signature & Date

Put the correct choice (a, b, c, d, or e) for each question in this table:

Question no.	1	2	3	4	5	6	7	8
Answer								
Question no.	9	10	11	12	13	14	15	Score/20
Answer								

Question one: (choose the correct answer for the following Questions from Q.1 to Q.10 and but the symbol of correct choice in the table in page one

في السؤال الأول المطلوب اختيار رمز الاجابة الصحيحة للأسئلة من سؤال 1 لسؤال 10 ووضع رمز الجواب الصحيح في الجدول بالصفحة الاولى

Q.1) Three students derive the following equations in which (x) refers to distance traveled, the (v) speed, (a) the acceleration, (t) the time, and the subscript zero means a quantity at time ($t = 0$). Here are their equations: (a) ($x = v_0 t + \frac{1}{2} a t^2$), and (b) ($x = v_0 t + 2 a t^2$) Which of these could possibly be correct according to a dimensional check?

A. a B. b C. both a and b are correct D. both a and b are not correct

Q.2) Estimate how many days it would take to walk around the Earth, assuming (12 h) walking per day at ($5 \frac{\text{km}}{\text{h}}$). (Hint: The radius of the Earth is about $R_{\text{Earth}} = 6380 \text{ km}$, and the distance around the Earth at the equator is the circumference, $2\pi R_{\text{Earth}}$)

A. 835 days B. 668 days C. 213 days D. 2403 days

Q.3) You drive 3 km at 30 km/h and then another 7 km at 50 km/h What is your average speed for the whole 10 km trip?

A. More than $40 \frac{\text{km}}{\text{h}}$ B. Equal to $40 \frac{\text{km}}{\text{h}}$ C. Less than $40 \frac{\text{km}}{\text{h}}$ D. Not enough information.

Q.4) How long does it take a car to cross a 15 m wide intersection after the light turns green, if the car accelerates from rest at a constant 2 m/s^2 ?

A. 7 s B. 3.87 s C. 5.48 s D. 15 s

Q.5) A person throws a ball upward into the air with an initial velocity of (7.5 m/s). How long is the ball in the air before it comes back to the hand? ($g = 9.80 \text{ m/s}^2$)

A. 3.06 s B. 4.87 s C. 1.48 s D. 1.53 s

Q.6) Ignoring air resistance, the horizontal component of a projectile's acceleration

A. is zero. B. remains a non-zero constant. C. continuously increases. D. continuously decreases.

Q.7) A stone is thrown horizontally with an initial speed of 10 m/s from the edge of a cliff. A stop watch measures the stone's trajectory time from the top of the cliff to the bottom to be 2.3 s.

What is the height of the cliff? ($g = 9.80 \text{ m/s}^2$)

A. 26 m B. 43 m C. 77 m D. 91 m

Q.8) Determine the direction of the resultant of the following three vectors with the x-axis (θ_x):

$$A = 3\hat{i} + 7\hat{j} + 8\hat{k}, B = 4\hat{i} - 5\hat{j} + 3\hat{k}, \text{ and } C = 2\hat{i} + 3\hat{j} - 4\hat{k}$$

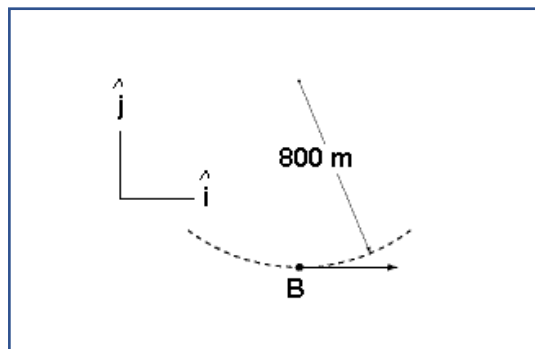
a. 55.8° b. 66.3° c. 43.7° d. 95°

Q.9) If $A = -3.75\hat{i} - 2.5\hat{j} + 1.5\hat{k}$ and, $= 32\hat{i} + 44\hat{j} + 64\hat{k}$, determine the magnitude of the vector C if ($C = A \times B$).

a. 85.7 b. 288 c. 375.8 d. 226

Q.10) A stunt pilot performs a circular dive of radius 800 m. At the bottom of the dive (point B in the figure) the pilot has a speed of 200 m/s which at that instant is increasing at a rate of 20 m/s^2 . What acceleration does the pilot have at point B?

- a. $(50\hat{i} + 20\hat{j}) \text{ m/s}^2$
- b. $(20\hat{i} - 50\hat{j}) \text{ m/s}^2$
- c. $(20\hat{i} + 50\hat{j}) \text{ m/s}^2$
- d. $(-20\hat{i} + 50\hat{j}) \text{ m/s}^2$
- e. $(-50\hat{i} + 20\hat{j}) \text{ m/s}^2$



Question Two: (Solve the following Questions from Q.11 to Q.15 with complete solution steps)

في السؤال الثاني المطلوب حل الأسئلة التالية من سؤال 11 لسؤال 15 حلاً كاملاً مع خطوات الحل

Q.11) A fly lands on one wall of a room. The lower-left corner of the wall is selected as the origin of a two-dimensional Cartesian coordinate system. If the fly is located at the point having coordinates (3, 2) m, what is its location in polar coordinates?

$$\begin{aligned}\theta &= \tan^{-1}(y/x) > \theta = \tan^{-1}(2/3.0) > \theta = 33.7 \\ x &= r \cos \theta > r = 3.00 / \cos 33.7 > r = 3.60\end{aligned}$$

Q.12) A person starts from point A, and walks due East at a speed of 10 m/s for 500 m, then stops for 60 seconds, after which he resumes the journey and travels further East for 2 minutes at 5 m/s to reach point B. Calculate, between points A and B: (i) the average speed and (ii) magnitude and direction of average velocity.

$$\begin{aligned}v_1 &= 10 \text{ m/s} & v_2 &= 5 \text{ m/s} \\ \Delta x_1 &= 500 \text{ m} & \Delta x_2 &= 5 \times 120 = 600 \text{ m} \\ t_1 &= \frac{500}{10} = 50 \text{ s} & t_2 &= 120 \text{ s}\end{aligned}$$

$$\text{i) Ave speed} = \frac{\text{Dist.}}{\text{time}} = \frac{500+600}{50+60+120} = \frac{1100 \text{ m}}{230 \text{ s}} = 4.78 \text{ m/s}$$

$$\text{ii) Velocity} = 4.78 \text{ m/s } \vec{E}$$

Q.13) If $A = 5i + 2j - 7k$, and $B = 2i + j - 8k$, find a) vector $C = 3A - 2B$ and b) the magnitude of C .

$$C = (15i + 6j - 21k) - (4i + 2j - 16k) = 11i + 4j - 5k \quad \text{Magnitude} = 12.7$$

Q.14) The velocity of a particle moving along the x axis is given for $t > 0$ by $v_x = 32t - 2t^3 \text{ m/s}$, where t is in s . What is the maximum velocity of the particle when (after $t = 0$) it achieves in the positive x direction?

Max velocity is when $a = 0$

$$0 = 32 - 6t^2$$

Gives $t = 2.3 \text{ sec}$

$$v = 32t - 2t^3$$

$$\text{At } t = 2.3 \text{ s, } v = 32(2.3) - 2(2.3)^3 = 49 \text{ m/s}$$

Q.15) A movie stunt driver on a motorcycle speeds horizontally off a 50 m high cliff (see figure below). How fast must the motorcycle leave the cliff top to land on level ground below, 90 m from the base of the cliff where the cameras are? Ignore air resistance. ($g = 9.80 \text{ m/s}^2$)

$$y = -\frac{1}{2}gt^2.$$

$$t = \sqrt{\frac{2y}{-g}} = \sqrt{\frac{2(-50.0 \text{ m})}{-9.80 \text{ m/s}^2}} = 3.19 \text{ s.}$$

$$x = v_{x0}t.$$

$$v_{x0} = \frac{x}{t} = \frac{90.0 \text{ m}}{3.19 \text{ s}} = 28.2 \text{ m/s,}$$

